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Supplementary information

Discrete confidence levels revealed by sequential decisions

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Supplementary Information

for ***"Discrete confidence levels revealed by sequential decisions"***, by

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1 Supplementary methods

1.1 Computational models

1.1.1 Ideal Bayesian observer

To model performance in the dual-decision task, we assumed that the observer has only access to r_i, a corrupted version of the true stimulus value s_i, defined as $r_i = s_i + \eta_i$, where the subscript $i = 1, 2$ indicate the decision at hand (first or second) and η_i is Gaussian noise with variance σ_i^2. Note that if the two decisions involve different perceptual judgments, the variance of the noise can be different depending on the decision (first or second). The perceptual discrimination task amounts to deciding whether s_i was above or below a reference level, call it zero. The ideal Bayesian observer has full knowledge of the statistics of the internal noise, and to make a decision computes the posterior probability that the variable s_i was positive or negative (i.e. above or below the reference level), given the noisy internal response r_i. Since the internal noise is Gaussian, the likelihood function giving the probability of observing r_i given s_i is a normal distribution with mean s_i and variance σ_i^2.

In the first decision, the prior probability of s_1 is uniform within the range $(-R, R)$, that is $p(s_1) = 1/(2R)$ if $-R \leq s \leq R$, and 0 otherwise. Combining the prior and the likelihood function, the unconditioned probability of observing r_1 can be expressed as:

$$\begin{aligned} p\left(r_1\right) &= \int_{-R}^{R} p\left(r_1|s_1\right)\ p\left(s_1\right)\ ds_1 \\ &= \frac{1}{4R}\left[\operatorname{erf}\left(\frac{R+r_1}{\sigma_1\sqrt{2}}\right) + \operatorname{erf}\left(\frac{R-r_1}{\sigma_1\sqrt{2}}\right)\right]. \end{aligned}$$

The posterior probability of s_1 after having observed r_1 is obtained applying Bayes rule:

$$p\left(s_1 \mid r_1\right) = \frac{p\left(r_1 \mid s_1\right) p\left(s_1\right)}{p\left(r_1\right)}.$$

Finally, the decision variable $c_1^+\left(r_1\right)$, corresponding to the probability that the s_1 was greater than 0 is given by

$$c_1^+\left(r_1\right) = \int_0^R p\left(s_1|r_1\right) ds_1 = \frac{\operatorname{erf}\left(\frac{r_1}{\sigma_1\sqrt{2}}\right) + \operatorname{erf}\left(\frac{R-r_1}{\sigma_1\sqrt{2}}\right)}{\operatorname{erf}\left(\frac{R+r_1}{\sigma_1\sqrt{2}}\right) + \operatorname{erf}\left(\frac{R-r_1}{\sigma_1\sqrt{2}}\right)}. \tag{1}$$

When $c_1^+\left(r_1\right) \geq 1/2$ the observer chooses 'right' (i.e. they decide to report that s_1 was positive, an outcome hereafter indicated with the notation d_1^+) otherwise they choose 'left' (i.e. they report that s_1 was negative, hereafter notated as d_1^-). The posterior probability of being correct in the first decision, which corresponds to the confidence of the ideal observer, is given by $c_1 = \max\left[c_1^+\left(r_1\right),\ 1 - c_1^+\left(r_1\right)\right]$. The ideal observer uses the probability c_1 to adjust prior expectations for the second decision, specifically by assigning a prior probability equal to c_1 to the possibility that s_2 will be drawn from the positive interval:

$$p\left(s_2\right) = \begin{cases} \frac{c_1}{R}, & \text{if } 0 < s_2 \leq R \\ \frac{1-c_1}{R}, & \text{if } -R \leq s_2 < 0 \\ 0, & \text{otherwise.} \end{cases}$$

By applying the same calculation as above with the updated prior $p(s_2)$, one obtains the decision variable $c_2^+(r_2)$ for the second decision:

$$c_2^+(r_2) = \frac{c_1\left[\mathrm{erf}\left(\frac{r_2}{\sigma_2\sqrt{2}}\right) + \mathrm{erf}\left(\frac{R-r_2}{\sigma_2\sqrt{2}}\right)\right]}{(1-c_1)\,\mathrm{erf}\left(\frac{R+r_2}{\sigma_2\sqrt{2}}\right) + c_1\mathrm{erf}\left(\frac{R-r_2}{\sigma_2\sqrt{2}}\right) + (2c_1-1)\,\mathrm{erf}\left(\frac{r_2}{\sigma_2\sqrt{2}}\right)}. \tag{2}$$

This equation reduces to the one for the first decision variable (apart from the difference in the variance of the noise) when $c_1 = 1/2$, as it should.

The range R on which s_1 and s_2 takes values is immaterial. It is possible to simplify Eqs (1) and (2) by taking the limit $R \to \infty$. In this limit one obtains:

$$c_1^+(r_1) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{r_1}{\sigma_1\sqrt{2}}\right)\right] \tag{3}$$

$$c_2^+(r_2) = \frac{c_1\left[1 + \mathrm{erf}\left(\frac{r_2}{\sigma_2\sqrt{2}}\right)\right]}{1 + (2c_1-1)\,\mathrm{erf}\left(\frac{r_2}{\sigma_2\sqrt{2}}\right)}.$$

The decision rules described by these equations amount to comparing the internal signal r_i to a criterion, and decide accordingly (i.e., if $r_i \geq \theta_2$, choose d_2^+, otherwise choose d_2^-). The criterion for the first decision, θ_1, can be expressed as:

$$c_1^+(\theta_1) = \frac{1}{2} \quad \Rightarrow \quad \mathrm{erf}\left(\frac{\theta_1}{\sigma_1\sqrt{2}}\right) = 0$$

which is satisfied for $\theta_1 = 0$. Similarly, the criterion for the second decision, θ_2, can be expressed as:

$$c_2^+(\theta_2) = \frac{1}{2} \quad \Rightarrow \quad \mathrm{erf}\left(\frac{\theta_2}{\sigma_2\sqrt{2}}\right) = 1 - 2c_1 \tag{4}$$

which indicates that θ_2 is a function of c_1 (see Main text, Figure 2B). Using Eqs (1) and (4), we can express θ_2 as a function of the internal response, r_1, as:

$$\theta_2 = -\frac{\sigma_2}{\sigma_1}\,|r_1|$$

Note that, if the noise is the same (i.e., $\sigma_1 = \sigma_2$), the optimal strategy amounts to shifting the criterion by the same amount as the internal response, that is $\theta_2 = -|r_1|$. Thus, when both decisions involve the same perceptual judgment, the optimal strategy does not necessarily imply the knowledge of the variability of the internal response. On the contrary, the optimal strategy does indeed require the knowledge of σ_1 and σ_2 (at least their ratio) when the decisions are made in different domains, as, e.g., in our experiment involving motion and orientation judgments (see Main text, Figure 1A).

The likelihood of the Bayesian observer choosing d_2^+ (after having chosen d_1^+ in the first decision) can be expressed as:

$$\begin{aligned} p\left(d_2^+ \,\middle|\, s_2, d_1^+, s_1\right) &= \int_0^{+\infty} p\left(d_2^+ \,\middle|\, \theta_2 = -\sigma_2 r_1/\sigma_1, s_2\right)\; p\left(r_1 \,\middle|\, d_1^+, s_1\right) dr_1 \\ &= \frac{1}{2} + \frac{1}{2p\left(d_1^+ \mid s_1\right)} \cdot \frac{1}{\sqrt{2\pi\sigma_1^2}} \int_0^{+\infty} \mathrm{erf}\left[\frac{1}{\sqrt{2}}\left(\frac{r_1}{\sigma_1} + \frac{s_2}{\sigma_2}\right)\right] \exp\left[-\left(\frac{r_1 - s_1}{\sigma_1\sqrt{2}}\right)^2\right] dr_1. \end{aligned} \tag{5}$$

If instead the observer has chosen d_1^- in the first decision the likelihood for the second decision takes the form:

$$p\left(d_2^+|s_2, d_1^-, s_1\right) = \int_{-\infty}^{0} p\left(d_2^+ \,\middle|\, \theta_2 = \sigma_2 r_1/\sigma_1, s_2\right)\; p\left(r_1 \,\middle|\, d_1^-, s_1\right) dr_1$$
$$= \frac{1}{2} - \frac{1}{2p\left(d_1^- \mid s_1\right)} \cdot \frac{1}{\sqrt{2\pi\sigma_1^2}} \int_{-\infty}^{0} \operatorname{erf}\left[\frac{1}{\sqrt{2}}\left(\frac{r_1}{\sigma_1} - \frac{s_2}{\sigma_2}\right)\right] \exp\left[-\left(\frac{r_1 - s_1}{\sigma_1\sqrt{2}}\right)^2\right] dr_1. \tag{6}$$

The integrals in the Eqs (5) and (6) were evaluated numerically using the adaptive quadrature algorithm as implemented in the function `integrate()` in **R** [10].

1.1.2 *Biased*-Bayesian observer

Similarly to the ideal Bayesian observer presented in the previous section, the biased-Bayesian observer also makes decisions based on internal signals corrupted by Gaussian noise. However, contrary to the ideal Bayesian observer, this model does not assume exact knowledge of the values of σ_1 and σ_2. In this case, the adjusted criterion for the second decision is given by

$$\theta_2 = -\frac{1}{m}\frac{\sigma_2\,|r_1|}{\sigma_1} = \frac{1}{m}\theta_2^{(opt)},$$

where $\theta_2^{(opt)}$ is the criterion of the Bayesian observer. When the ratio of the variances of the internal noises is overestimated ($m > 1$), the criterion is shifted away from zero less than in the optimal Bayesian model. This leads to an under-confidence bias, i.e., in the second decision the observer chooses *as if* the first decision was correct more frequently than expected if their performance was optimal. Conversely, when the ratio of the variances of the internal noises is underestimated ($m < 1$), the criterion is shifted more than in the optimal Bayesian model, leading to an opposite over-confidence bias.

1.1.3 *Discrete* confidence model

As an alternative to the optimal Bayesian model we considered a class of models that do not assume any knowledge about the nature of the internal stochastic process linking the stimulus s_i with the internal observation r_i. These non-Bayesian observers perform similarly to the Bayesian model for the first decision, that is when $r_1 \geq 0$ they choose d_1^+, and d_1^- otherwise. However, they cannot estimate a full probability distribution over the values of s_i, and therefore can assess confidence only by comparing the internal response r_i - which can be described as a point-estimate - to a set of one (or possibly more) fixed criteria. In the case of a single confidence criterion, this non Bayesian observer is in a high-confidence state when the internal signal exceeds the confidence criterion, and uncertain otherwise. When confident about the first response, the observer shifts the criterion for the second decision by a fixed amount (thereby increasing the probability of choosing d_2^+). If only one criterion is used the model has 2 discrete confidence levels (e.g., 'uncertain' vs 'confident'). In such a 'discrete' model, if the confidence criterion is w_1, the probability of the observer being confident about their first decision, after having responded d_1^+, can be expressed as:

$$p\left(\text{confident} \,\middle|\, s_1, d_1^+\right) = \frac{1 - \Phi\left(\frac{s_1 - w_1}{\sigma_1}\right)}{1 - \Phi\left(\frac{-w_1}{\sigma_1}\right)}$$

where Φ is the cumulative distribution function of the standard normal distribution. Note that this probability does not denote the confidence of the observer about their choice, which instead is assumed here to be a discrete binary state. Applying the law of total probability, the probability of the observer reporting d_2^+ in the second decision can be expressed as:

$$p\left(d_2^+ \,\middle|\, s_2\right) = p\left(\text{confident} \,\middle|\, s_1, d_1^+\right) \Phi\left(\frac{s_2 - \theta_2}{\sigma_2}\right) + \left[1 - p\left(\text{confident} \,\middle|\, s_1, d_1^+\right)\right] \Phi\left(\frac{s_2}{\sigma_2}\right)$$

where θ_2 is the shift in criterion for the second decision applied by the observer when they are confident in their first decision. w_1 and θ_2 are free parameters that we fit to the data by maximum likelihood estimation. It is straightforward to extend the model in order to have more than two confidence levels, however we find that increasing the number of additional levels in the model provides, in the current dataset, only a negligible increase in likelihood which is not sufficient to justify the additional complexity. The difference in AIC score between the simple model with only 2 confidence levels outlined above and a model with one additional confidence criterion (and therefore 3 total confidence levels) was 102 ± 5 (summed over subjects) favouring the simpler model with only 2 confidence levels.

1.2 Sampling-based approximation of Bayesian observer

An interesting alternative to the models presented in the previous sections is represented by models where the observer does not have access to the full probability distribution of their internal signals, but bases their decision on a limited number of samples. In these models the posterior probability that the choice is correct (the confidence) is approximated on the basis of a fixed number n of samples $x_1, x_2, ..., x_n$ drawn from the posterior distribution $p(s_i \mid r_i)$. The performance of these sampling-based models will approach the optimal Bayesian model as $n \to \infty$, however they are expected to display systematic biases and deviations from the optimal model for a small number of samples [12]. Here we show that the sampling-based approximation of the Bayesian observer will display a systematic over-confidence bias, that is in the opposite direction to that displayed by most participants in our study, and is thus inconsistent with the behavioural pattern found in our data.

1.2.1 Sampling bias and over-confidence

It has been shown that when a probability p is estimated from a small sample, as the empirical frequency of successes k out of n random trials, $\hat{p} = k/n$, the risk of over-estimation (that is $\hat{p} > p$) or under-estimation, ($\hat{p} < p$) depends in a complex way on both the probability p and the sample size n [13]. This is however for estimating a single, fixed probability p. What would instead be the average (expected) bias, over many repeated estimations, when the probability p varies randomly within a given range? In our experiments confidence is the posterior probability that a binary choice is correct, and as such it varies from complete uncertainty, $p = 0.5$, to complete certainty, $p = 1$, hence $p \in [0.5, 1]$. We show here that when a set of probabilities $p_1, ..., p_m$ uniformly distributed in the interval $[0.5, 1]$ is estimated using a limited number of samples n, the expected bias is one of over-estimation.
For a given n and p the probabilities of over- and under- estimation can be expressed as:

$$p\left(\hat{p} > p | n, p\right) = \sum_{k=\lfloor np \rfloor + 1}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

$$p\left(\hat{p} < p|n,p\right) = \sum_{k=1}^{\lceil np \rceil + 1} \binom{n}{k} p^k (1-p)^{n-k}$$

where $\lceil . \rceil$ and $\lfloor . \rfloor$ are the ceiling and the floor operators, i.e. functions that map a number to the smallest following integer or the largest previous integer, respectively. Following Shteingart and Loewenstein [13] we consider the difference between these two, denoted as *probability estimation bias*, which takes the form:

$$\Delta p = p\left(\hat{p} > p|n,p\right) - p\left(\hat{p} < p|n,p\right).$$

When Δp is positive, it indicates that the probability p is more likely to be overestimated than underestimated, and vice versa for negative values. Assuming that all values of p in the interval are equally likely, the expected bias can be computed by integrating Δp over the range of p (that is $[0.5, 1]$):

$$\mathbb{E}\left[\Delta p\right] = \frac{1}{2} \int_{0.5}^{1} \Delta p \, dp.$$

A positive value of the expected probability estimation bias (that is $\mathbb{E}\left[\Delta p\right] > 0$) indicates that, on average, the probabilities in this interval are more frequently overestimated rather than underestimated. This integral can be evaluated numerically, and in Supplementary Figure 1A we plotted the expected probability estimation bias as a function of the number of samples, for two different ranges. When p varies within the range of confidence, $[0.5, 1]$, the value of the probability estimation bias is always positive, although modulated by the number of samples, indicating that in the range $[0.5, 1]$ overestimation is more likely than underestimation.

1.2.2 Fixed-n Bayesian sampler

Here we present the details of a model based on a sampling approximation of the Bayesian observer, and confirm by simulation that it will results in an over-confidence bias. We consider a fixed-n policy, where the observer draws a fixed number of samples for each decision. Although alternative decision policies are possible (such as an accumulator policy, where the decision is taken after a minimum number of samples is accumulated in favour of one of the options), these have been shown elsewhere to result in very similar performances as the fixed-n policy [15].

Similarly to the other models, we assume that the observer has only access to r_1, a corrupted version of the stimulus s_1, $r_1 = s_1 + \eta_1$, where η_1 is Gaussian noise with variance σ_1^2. In the first decision the prior is flat and, taking the limit of the stimuli range $R \to \infty$, the posterior distribution $p\left(s_1|r_1\right)$ results in a Gaussian distribution centred on the internal observation r_1. The probability that a sample from this distribution is above 0 (the criterion for the first decision) can be computed as:

$$p\left(x_n > 0|r_1\right) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{r_1}{\sigma_1\sqrt{2}}\right)\right] = \Phi\left(\frac{r_1}{\sigma_1}\right).$$

The conditional probability that the observer chooses d_1^+ given r_1 is obtained by summing the probability of all the set of samples with at least $\lceil \frac{n}{2} \rceil$ samples above 0 and can be expressed as:

$$p\left(d_1^+ \,\middle|\, r_1\right) = \sum_{k=\lceil \frac{n}{2} \rceil}^{n} \binom{n}{k} \Phi\left(\frac{r_1}{\sigma_1}\right)^k \left[1 - \Phi\left(\frac{r_1}{\sigma_1}\right)\right]^{n-k}.$$

In other words the observer choses d_1^+ when the majority of samples is above 0. The observer's confidence in his decision, $c_1 = \max\left[c_1^+, 1 - c_1^+\right]$, is given by the proportion of samples in favor of the choice made:

$$c^+ = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{x_i>0} \tag{7}$$

where $\mathbb{1}_{x_i>0}$ denotes the indicator function ($\mathbb{1}_{x>0} = 1$ if $x > 0$, and 0 otherwise). The probability that the observer chooses d_1^+ given s_1 can be calculated by integrating the probability $p\left(d_1^+ \,\middle|\, r_1\right)$ over all possible values of r_1:

$$p\left(d_1^+ \,\middle|\, s_1\right) = \sum_{k=\lceil \frac{n}{2} \rceil}^{n} \binom{n}{k} \int_{-\infty}^{\infty} \Phi\left(\frac{r_1}{\sigma_1}\right)^k \left[1 - \Phi\left(\frac{r_1}{\sigma_1}\right)\right]^{n-k} p\left(r_1 \mid s_1\right) dr_1. \tag{8}$$

In the second decision the prior probability of the stimulus s_2 is different for stimuli above or below 0. From the point of view of the observer, the prior probability that the stimulus for the second decision s_2 is above 0 corresponds to the confidence c_1 that the first decision was correct and, conversely, $p\left(s_2 < 0\right) = 1 - c_1$. Updating the prior probability for the second decision amounts to a shift in the decision criterion, as demonstrated for the full Bayesian model. The shift in criterion for the second decision θ_2 is a function of the confidence in the sampling model as in the full Bayesian model

$$\theta_2 = \sigma_2\sqrt{2}\ \mathrm{erf}^{-1}\left(1 - 2\mathrm{c}_1\right). \tag{9}$$

However, and differently from the full Bayesian model, the confidence is limited to a finite set of values determined by the number of samples n. The number of possible confidence levels is $\left\lceil \frac{n+1}{2} \right\rceil$, and each of these corresponds to a value of θ_2. If the observer's first decision was d_1^+, then the probability of the confidence level c_j

$$p\left(c_j \,\middle|\, s_1, d_1^+\right) = \binom{n}{k} \int_{-\infty}^{\infty} \Phi\left(\frac{r_1}{\sigma_1}\right)^k \left[1 - \Phi\left(\frac{r_1}{\sigma_1}\right)\right]^{n-k} p\left(r_1|s_1\right) dr_1$$

where $j \in \left(1, ..., \left\lceil \frac{n+1}{2} \right\rceil\right)$ is an index linked to the number of samples above the criterion, k, according to $k = \left\lfloor \frac{n-1}{2} \right\rfloor + j$. Conversely, if the observer's first decision was instead d_1^-, then the probability of the confidence level c_j is

$$p\left(c_j|s_1, d_1^-\right) = \binom{n}{k} \int_{-\infty}^{\infty} \left\{1 - \Phi\left(\frac{r_1}{\sigma_1}\right)^k \left[1 - \Phi\left(\frac{r_1}{\sigma_1}\right)\right]^{n-k}\right\} p\left(r_1|s_1\right) dr_1 \tag{10}$$

and the relation between the index of confidence level j and the number of samples above 0, k, becomes $k = n - \left(\left\lfloor \frac{n-1}{2} \right\rfloor + j\right)$.
A level of confidence c_j would result in a shift in decision criterion θ_j, calculated according to Eq (9). The probability of choosing d_2^+ in the second decision is

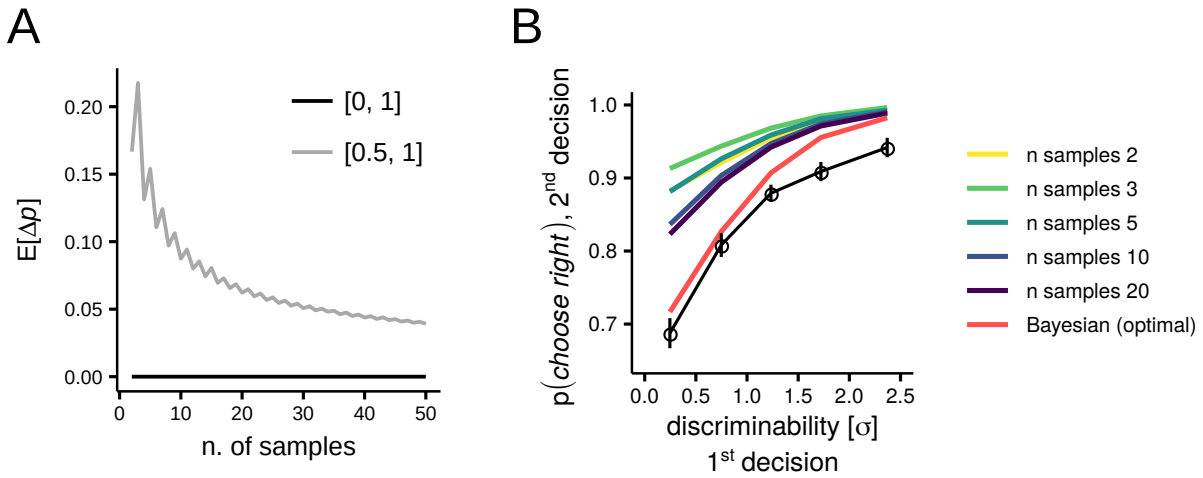
$$p\left(d_2^+ \,\middle|\, s_2, s_1, c_j\right) = \sum_{k=\lceil \frac{n}{2} \rceil}^{n} \binom{n}{k} \int_{-\infty}^{\infty} \Phi\left(\frac{r_2 - \theta_j}{\sigma_2}\right)^k \left[1 - \Phi\left(\frac{r_2 - \theta_j}{\sigma_2}\right)\right]^{n-k} p\left(r_2|s_2\right) dr_2.$$

Taking everything together, the probability of the observer choosing d_2^+ given s_2, s_1 and the first decision can be expressed as:

$$p\left(d_2^+ \;\middle|\; s_2, d_1^+, s_1\right) = \sum_{j=1}^{\lceil \frac{n+1}{2} \rceil} p\left(c_j | s_1, d_1^+\right)\; p\left(d_2^+ \;\middle|\; s_2, s_1, c_j\right).$$

1.2.3 Simulation of sampling model

We simulated the model for values of n ranging from 2 to 20. In order to compare the model with the full Bayesian observer, the values of σ for each n are adjusted so as to obtain the same proportion of correct responses in the first decision. For each value of n and for 100 iterations we first estimated the σ for the sampling models based on the set of first decisions made by the participants (this was done using maximum likelihood estimation and equation 8) and simulated second decisions from the sampling model, conditioned on the actual set of observed first decisions made by the participants. The proportion of responses 'right' (d_2^+) in the second decision produced by the sampling models are plotted in Supplementary Figure 1C. As expected, they show a pattern of marked over-confidence: all sampling models tended to respond d_2^+ more often than the optimal model, despite similar accuracy in the first decision. The bias is generally larger for models with smaller number of samples, and decreases approaching the optimal Bayesian model as n increases. Importantly, this bias is incompatible with the observed behavioral data, which instead showed a marked under-confidence bias.



Supplementary Figure 1: Sampling-based approximation of the Bayesian model. A. Expected probability estimation bias Δp plotted as a function of the number of samples when the probability p varies randomly (uniformly) either in the range $[0.5, 1]$, plotted in grey, or in the range $[0, 1]$. It can be seen that when p varies within the range of confidence, from chance to certainty $[0.5, 1]$ the predominant bias is one of over-estimation (because Δp is always positive, grey line). Only when p varis over the whole domain of probability, $[0, 1]$, the expected bias is on average zero and over-estimation and under-estimation are equally likely (black line). **B**. Proportions of responses 'right' (d_2^+) in the second decision as a function of the difficulty of the first decision, as predicted by the full Bayesian model (red line) and the fixed-n Bayesian sampler model. The black dots and lines indicate the choices made by human participants (error bars are bootstrapped standard errors).

1.3 Model estimation and comparison

In our analyses we compared each model's ability to explain performance in the second decision. The first step was to estimate the standard deviation of the internal noise, using only decisions where the prior probability was equal for the two alternatives (that is first decisions and control conditions). In fact in the first decision s_1 is uniformly chosen from an interval centred around 0 (i.e., s_1 will be above 0 with probability 1/2), and there is no prior information about the sign of s_1. Therefore, for all the models presented in the previous sections (with the exception of the sampling approximation of the ideal Bayesian observer discussed in the previous section), the probability $p\left(d_1^+ \mid s_1\right)$ of the observer choosing d_1^+ can be expressed as:

$$p\left(d_1^+ \,\middle|\, s_1\right) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{s_1}{\sigma_1\sqrt{2}}\right)\right] = \Phi\left(\frac{s_1}{\sigma_1}\right).$$

We estimated by maximum likelihood $\hat{\sigma}_1$ for each participant and task. We also estimated other psychometric models, that extended this simple model to account for the possibility that the observer was biased, $p(d_1^+ \mid s_1) = \Phi\left(\frac{s_1-b}{\sigma_1}\right)$ (where b indicates the bias), or made stimulus-independent errors (e.g. attention lapses) with non-zero probability, $p(d_1^+ \mid s_1) = \lambda + (1-2\lambda)\Phi\left(\frac{s_1}{\sigma_1}\right)$ (where λ is the lapse probability); or both $p(d_1^+ \mid s_1) = \lambda + (1-2\lambda)\Phi\left(\frac{s_1-b}{\sigma_1}\right)$. Since we were interested in estimating the internal noise, rather than selecting one of these models, we averaged the estimates of $\hat{\sigma}$ obtained with each of these psychometric models according to their Akaike weights [1, 16]. In more detail, we first calculated the Akaike Information criterion (AIC) of each psychometric models m as

$$\mathrm{AIC}_m = 2\mathcal{V} - 2\ln\hat{\mathcal{L}}_m$$

where $\hat{\mathcal{L}}_m$ is the maximized value of the likelihood function of model m and $\mathcal{V}$ its number of free parameters. The AIC values are first transformed into differences with respect to the best model, $\Delta_m = \mathrm{AIC}_m - \min \mathrm{AIC}$. These differences are then transformed on a likelihood scale and normalized to obtain the Akaike weights w_m, which can be directly interpreted as conditional probabilities for each model [16]

$$w_m = \frac{\exp\left(-\frac{1}{2}\Delta_m\right)}{\sum\limits_{k=1}^{K}\exp\left(-\frac{1}{2}\Delta_k\right)}.$$

Finally, the model-averaged estimate of the noise parameter is computed as the average of each estimated weighted by the Akaike weight of the respective models

$$\bar{\sigma}_1 = \sum_{k=1}^{K} w_k \hat{\sigma}_1^{(k)}.$$

The advantage of this approach is that it avoids the need to choose and commit to a particular form of the psychometric function. Furthermore, because the Akaike weights take into account the complexity of the psychometric functions (i.e. the number of free parameters $\mathcal{V}$), the resulting estimate optimises the balance between model fidelity to the data and the need to avoid overfitting (i.e. of fitting a model that is too complex to be supported by the data). Indeed the model-averaged estimate is more likely to perform better than the alternatives in accounting for new data [1].

In our analysis we used the model-averaged estimate for transforming the stimuli from physical units to units of internal noise. This simplifies the analysis (since first and second decisions involved different perceptual judgements) and guarantees that any differences between models would be due only to differences in explaining performance in the second decision. The same was done for the estimates of the bias term, $\hat{b}$, which was then subtracted from the stimuli value to take into account subjective decision biases in the computation of models' likelihoods. The remaining free parameters for the biased-Bayesian and the non-Bayesian observer were estimated by numerically maximising the (log) likelihood of the data (the observed patterns of second decisions). Maximum likelihood fits were obtained via the BOBYQA algorithm [9], implemented in the optimx package [7, 8] in **R** [10]. The relative quality of models' fit was assessed with the AIC, which we report pooled over subject in the main text; individual differences in AIC values are represented in Extended Data Figure 1.

In addition to the AIC, we compared the models using a random-effects approach [11, 14] which allows for the possibility that different participants could have used different strategies (corresponding to different models) to generate their behaviour. In this approach models are treated as random effects that could differ between subjects and have a fixed (unknown) distribution in the population. The uncertainty around model frequencies is described by a Dirichlet distribution (a multivariate generalisation of the Beta distribution), whose parameters are estimated according to an iterative variational Bayesian procedure, described in [14][1], starting from a measure of the evidence supporting each model (here we used the AIC). Having obtained the optimised parameters of the Dirichlet distribution a useful way to rank the model is according to their exceedance probability, that is the second-order probability that participants were more likely to choose a certain model to generate their behaviour rather than any other alternative model. If $r_1, \ldots, r_K$ are the estimated frequencies of the K models, the exceedance probability (EP) of model k is formally defined as $EP_k = p\left(r_k > r_j \mid \text{data}\right)$, with $k \neq j$. Since we are considering more than two models, the exceedance probabilities need to be approximated via simulation [14]. We simulated 10^6 multivariate Dirichlet samples $\tilde{r}_1, \ldots, \tilde{r}_K$ and calculated the exceedance probability of model k as

$$EP_k = \frac{\sum \mathbb{1}_{\tilde{r}_k > \tilde{r}_j}}{10^6}, \forall j \in \{1, \ldots, K, j \neq k\}$$

where $\mathbb{1}$ is the indicator function ($\mathbb{1}_{x>0} = 1$ if $x > 0$ and 0 otherwise), summed over the total number of multivariate samples drawn. *Protected* exceedance probabilities (PEP) [11] are a further extension of exceedance probabilities corrected for the possibility that the observed model differences are due to chance. They are calculated as $PEP_k = (1 - \text{Bor})\, EP_k + \text{Bor}/K$, where Bor is the Bayesian omnibus risk, that is the posterior probability that all models have the same frequency [11]. See Main Text and Figure 3B for a plot of the estimated model frequencies with the associated exceedance probabilities and Bayesian omnibus risk.

1.4 Model-agnostic analyses

We report here the details of analyses that did not involve fitting the decision models presented in Section 1.1. We analysed performance by means of a series of logistic regressions. For the analysis of response accuracy, we estimated logistic regression of the form

$$\text{logit}\left[p(\text{correct})\right] = \beta_0 + \beta_1 \left|s\right| + \beta_2\, \text{DecisionOrder} + \beta_3\, \text{Dual} + \beta_4\, (\text{Dual} \times \text{DecisionOrder}) \quad (11)$$

[1] Implemented in an **R** package available at `http://github.com/mattelisi/bmsR`

where $\text{logit}\,[p] = \log \frac{p}{1-p}$, $|s|$ is the absoluted value of the stimulus (i.e. the evidence discriminability, expressed in units of internal noise, σ) and DecisionOrder (first vs second) is a dummy variable that was set to 1 in the case of second decisions and 0 otherwise. Dual was also a dummy variable, set to 1 for the dual-decision condition, and 0 for the control condition. In the main text we report 95% CI on the odds-ratio of a correct response in the second decision relative to that in the first decision; for example, in equation (11) this would be calculated as e^{β_2} for the control condition and as $e^{\beta_2+\beta_4}$ for the dual-decision condition.

We performed further analyses to assess the possibility that the sub-optimality of human behaviour could be explained by a fixed-bias model which does not adjust decision criterion dynamically depending on the stimulus observed at the first decision. We analysed the proportion of 'right' alternatives chosen in the second decision as a function of the stimulus value in the second decision and the accuracy of the first decision

$$\text{logit}\,[p(\text{choose right})] = \beta_0 + \beta_1 s_2 + \beta_2\,\text{Acc}_1$$

where Acc_1 is a dummy variable set to 1 in the case of correct first decisions. This logistic regression showed that participants were more likely to choose 'right' after correct first decisions (see Main Text). This logistic regression was also extended to include the interaction between accuracy and evidence discriminability in the first decision, that is

$$\text{logit}\,[p(\text{choose right})] = \beta_0 + \beta_1 s_2 + \beta_2\,\text{Acc}_1 + \beta_3 s_1 + \beta_4\,(|s_1| \times \text{Acc}_1)\,.$$

This analysis showed that the proportion of 'right' choices (in the second decision) significantly increased or decreased with the discriminability of the first stimulus ($|s_1|$) depending on whether the first decision was correct or wrong (see Main Text and Figure 1B).

2 Supplementary results

2.1 Analysis of response times

We conducted additional analyses to investigate how the response time (RT, an approximation of the deliberation time required to form the decision) in the second decision was related to the confidence in the first decision. Note that this analysis is not aimed at discriminating between models. In fact our models do not consider the dynamical aspects of the decision process, and do not make specific predictions about how degree of confidence and speed of decision should be related. Moreover, in our experiments we explicitly asked participants to prioritise accuracy, without worrying about the speed of their responses. Nevertheless, it is known that the confidence in a decision is usually inversely correlated with the response time [5]. Our goal here is therefore to provide a description of the relationship between confidence and speed of response in our dual decision paradigm, in order to provide a more complete description of the data and inspire future studies on the relationship between speed and confidence in the context of sequential decisions. The rationale of the analysis is the following: we begin by partialling out the effect of stimulus discriminability (both first and second) on the RT of the second decision. Next we calculate the expected confidence of the first decision (according to the ideal-Bayesian model, which corresponds to the objective probability of a correct response) and examine whether and how the expected confidence can account for the residual variations in RTs.

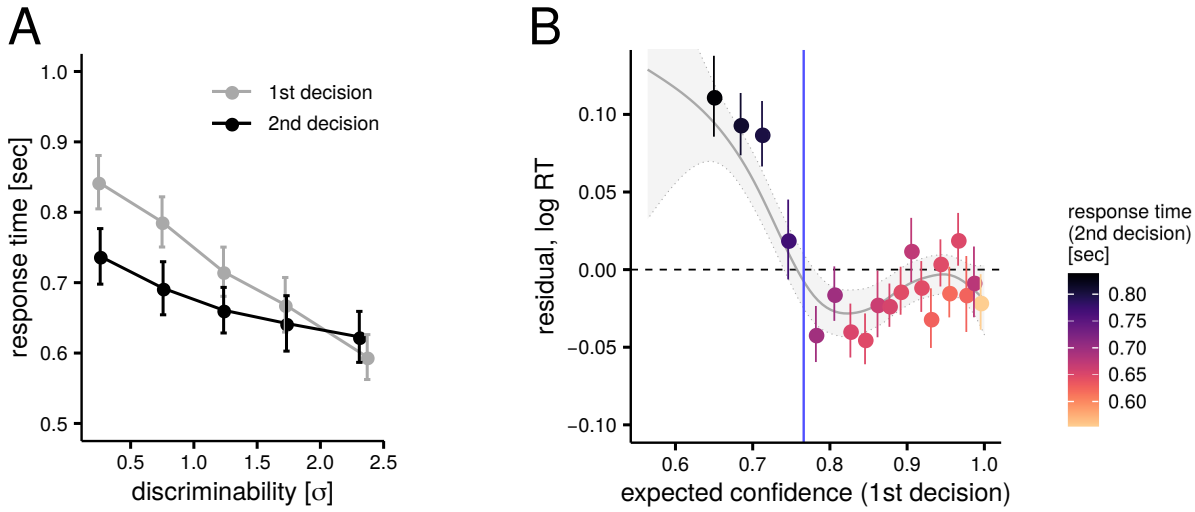
Given that participants had as much time as they wanted to respond, the RT distributions are likely to be contaminated by outliers. Therefore, prior to this analysis we removed trials with RTs longer than 5 sec after the offset of the stimulus ($\approx 0.28\%$ of total trials). The average RT (measured from the onset of the stimulus) in the first decision was 0.74 sec (SD across participants 0.19 sec), slightly slower than that of the second decision, 0.68 sec (SD 0.19 sec), $t(28) = 2.73$, $p = 0.01$ (see Supplementary Figure 2A). This difference was found only in the dual-decision condition: in the control condition RTs in the first and second decision had similar duration: first decision, mean 0.80 sec, SD 0.23 sec; second decision, mean 0.81 sec, SD 0.22 sec; $t(28) = 0.57$, $p = 0.57$, the Bayes factor (null/alternative) was 5.96, providing moderate support for the null hypothesis.

To control for the effect of stimulus discriminability on RTs, we fit a linear model to the logarithm of RTs, which included as predictors both the stimuli of first and second decisions as well as their interactions. Furthermore, given that RTs also tend to differ between correct and wrong responses, we also included the accuracy of first and second decisions as predictors. The regression model thus had the form

$$\log\left(\text{ResponseTime}\right) = \beta_0 + \beta_1 \left|s_1\right| + \beta_2 \left|s_2\right| + \beta_3 \left|s_1\right| \times \left|s_2\right| + \beta_4 \text{Acc}_1 + \beta_5 \text{Acc}_2 + \epsilon$$

where $\left|s_1\right|, \left|s_2\right|$ denote the absolute values of the stimuli in first and second decisions (the evidence discriminability), Acc_1 and Acc_2 the accuracy of first and second decisions, and ϵ a normally distributed residual error. The model was fit individually for each participant. The residuals of this model (i.e. the trial-by-trial differences between the observed and predicted logarithm of RTs) represent the residual variation in response times that was not accounted for by the stimuli levels and response accuracy.

Next, we computed the expected confidence according to the ideal observer. We do not have access to the noisy internal response of the participants (r_1), however we can calculate its expected value given the entire sequence of stimuli (s_1, s_2) and decisions (d_1, d_2). The distribution of the noisy internal response is a Gaussian centered on the true stimulus value. After observing the first decision of the participants, we can infer on which side of the decision criterion the internal response fell.



Supplementary Figure 2: Analysis of response times. A. Average response times in the dual-decision task, split by decision order and plotted as a function of the stimulus value (in units of internal noise). This plot shows that response times tend to be faster in the second decision than in the first, and particularly so for difficult decisions, mimicking the effect found in response accuracies (see Figure 1, Main text). Furthermore, this plot shows that response times decreases approximately linearly with the discriminability of the stimulus (i.e. with the average distance of the stimulus from the decision criterion). **B**. The residual variation in the response times of the second decision (after partialling out the effects of the two stimuli level and response accuracies, see text for details) is plotted as a function of the expected confidence of the Bayesian model in the first decision. Each point is the average residual (in logarithmic units) across participants for a given interval of confidence levels; error bars represent standard error of the mean. Bins are chosen such that each bin contains approximately the same number of trials. The smooth grey line represents a smooth GAM fit to the unbinned residuals (with 95% confidence interval). The color of each dot reprents the average response time in that bin, in seconds. Finally, the vertical blu line is the confidence level corresponding to the location at which participants placed the confidence criterion, according to the discrete model.

Therefore, the probability density function of the internal responses becomes a truncated Gaussian distribution. For example, when the decision is d_1^+, the distribution of r_1 can be expressed as

$$p\left(r_1 \mid d_1^+, s_1\right) = \frac{\phi\left(\frac{r_1 - s_1}{\sigma_1}\right)}{\sigma_1\left(1 - \Phi\left(\frac{-s_1}{\sigma_1}\right)\right)}$$

which is a truncated Gaussian bounded from below at 0. (Φ and ϕ are the cumulative distribution function and the probability density function of the standard normal distribution, respectively.) The distribution of the internal response is then obtained by applying Bayes theorem

$$p\left(r_1 \mid d_1, s_1, d_2, s_2\right) = \frac{p\left(d_2 \mid s_2, r_1\right) \;\; p\left(r_1 \mid d_1, s_1\right)}{\int\limits_{-\infty}^{\infty} p\left(d_2 \mid s_2, r_1\right) p\left(r_1 \mid d_1, s_1\right) dr_1}.$$

For each trial, we computed the expected value of the internal response, that is $\mathbb{E}\left[r_1 \mid d_1, d_2, s_1, s_2\right] = \int_{-\infty}^{\infty} r_1 \; p\left(r_1 \mid d_1, s_1, d_2, s_2\right) dr_1$, and plugged it into equation (3) to obtain the expected confidence level.

We found that the expected confidence level (*in the first decision*) can explain the residual variation of RTs *in the second decision*. Specifically, second decisions are faster or slower - relative to the conditional mean of the log RT predicted by the regression model - when confidence in

the first decision was higher or lower, respectively, as shown in Supplementary Figure 2B. Thus, this findings indicates that dynamical aspects of the second decision were adjusted based on trial-by-trial confidence in the first decision. Furthermore, we find that the RT residuals do not seem to decrease linearly with confidence, but rather are above the marginal mean when confidence is below a certain level (approximately corresponding to the average confidence level at which participants placed their confidence criterion w_1, as estimated by the discrete model, ≈ 0.77) then decrease rapidly below the conditional mean when confidence exceeds this level. This suggests a possible chronometric correlate of the discrete confidence states that we inferred from modelling participants' decisions. Although serendipitous, we believe this is an interesting result that could inspire future studies to apply dynamical models of decision making (e.g. drift diffusion models) in a dual-decision setting.

2.2 Dual-decision within the same perceptual task

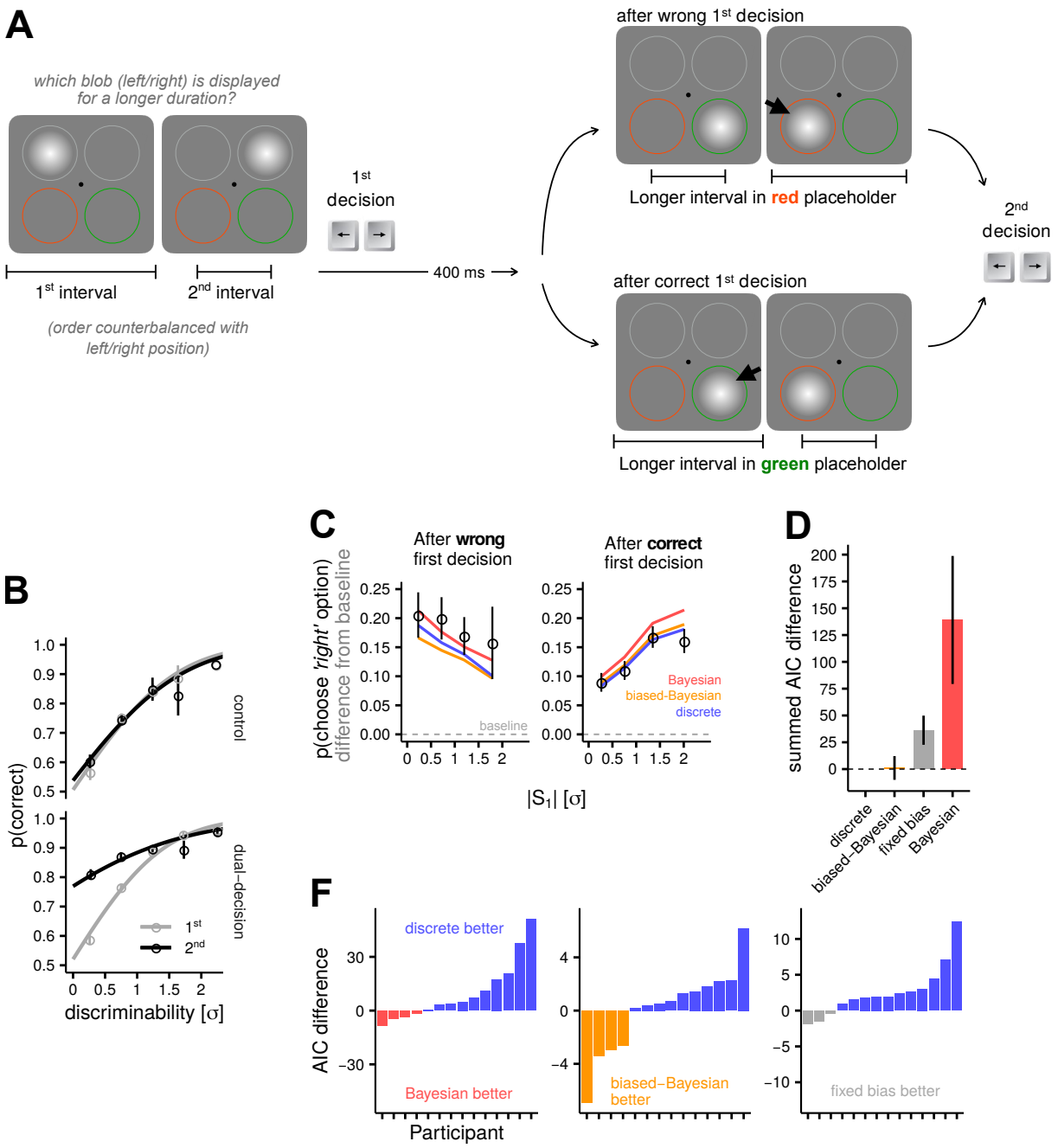
Here we report the results of an additional experiment in which both decisions involved the same perceptual task, specifically a judgment about the duration of a visual stimulus (Supplementary Figure 3A). Although overall the results show a pattern similar to that found in our main 'crossed-task' experiment, they are also suggestive of slightly more efficient use of confidence information when the two decisions involve the same ('within-task'), as opposed to different ('crossed-task'), perceptual judgments. At first sight, a benefit in within-task relative to crossed-task experimental designs would seem to challenge the notion of confidence as a common currency [2, 3] (i.e. represented on a common scale) between different perceptual judgments. However, as explained below, within-task designs present some limitations. This, combined with differences in procedure across the twp experiments, makes strong conclusions based unwarranted.

One issue with within-task designs is that they are less constrained in terms of the possible strategies adopted by participants and, as a result, are less informative about the nature of the probability representations underlying performance in the dual decision task. To see that, recall the relationship that we found between the internal response r_1, elicited by the first signal and the optimal criterion θ_2 for the second decision, i.e.,

$$\theta_2 = -\frac{\sigma_2}{\sigma_1} |r_1| . \tag{12}$$

If the two decisions involve the *same* preceptual judgement, it is not unreasonable to assume that $\sigma_1 = \sigma_2$, and then $\theta_2 = -|r_1|$. An observer aware of this mapping could achieve optimal performance having available only a point-estimate of the first signal, without needing any knowledge about the internal noise. Such an observer would be indistiguishable from the optimal Bayesian model whilst using an entirely non-probabilistic strategy [6].

Our main crossed-task experiment instead used 2 different perceptual judgments. Since the signals upon which these judgments are based are different in nature (mean tilt from vertical versus signed motion coherence), they are processed by distinct neural mechanisms and likely encoded in different internal representations. Optimal performance in this task requires comparing the uncertainty in these two representations, and therefore implies some knowledge about their noise statistics (at the very least the ratio between the internal noises in the two tasks, σ_1 and σ_2, since the optimal mapping becomes $\theta_2 = -\frac{\sigma_2}{\sigma_1} \times |r_1|$). Our results from the crossed-task experiment indicate that participants do not use this mapping, but rather seem to rely on the simpler strategy of shifting



Supplementary Figure 3: Dual-decision within the same perceptual task. **A**. Experimental protocol of the within-task experiment on duration judgments. **B**. Probability of correct first and second decisions in control and dual-decision condition; error bars are bootstrapped standard errors across observers, and smooth lines are probit functions fit to the aggregated data. **C**. Probability of choosing 'right' in the second decision broken down by the accuracy of the first response. **D**. Difference in AIC relative to the discrete confidence model, summed over subjects; error bars are boostrapped standard errors. **E**. Ranked individual AIC differences.

the second decision's criterion by a fixed amount whenever their point-estimate of the first signal exceed a fixed confidence criterion (i.e. the discrete model). In the case of a within-task design, however, participants may use the trial-by-trial point-estimate of the first signal (already available to them) directly as criterion in the second decision. Furthermore, they could use this optimal (but heuristic) mapping only in a fraction of trials while falling back on the simpler strategy in other trials. This creates an interpretational confound that complicate disentangling if performance in the within-task case is truly underpinned by a Bayesian decision strategy.

2.2.1 Methods

The experiment was run at the Université Paris Descartes, and the protocol of the experiments was approved by the Paris Descartes University Ethics Committee for Non-invasive Research (CERES). A total of 14 in Experiment 2 (mean age 33, standard deviation 7; 6 females, 2 authors) participated in the experiment. An additional group of 5 subjects participated in the control condition (mean age 27, standard deviation 3; 3 females). All participants had normal or corrected-to-normal vision and gave their informed consent in written form prior to participation. All participants (except the two authors) were naïve to the specific purpose of the experiment. Naïve participants were compensated with either 10€ for each hour of experiment.

The visual display of the within-task design consisted of a central fixation point and four placeholders, Supplementary Figure 3A, continuously visible on a uniform grey background. The four placeholders were circles (anti-aliased) measuring 2.8 dva in diameter, whose centers were placed at 1.8 dva from the fixation point. Two placeholders were light grey, and were placed above the horizontal mid-line; the other two were placed below the mid-line and were colored in red the one on the left, and in green the one on the right. Stimuli consisted of white Gaussian blobs (standard deviation 0.65 deg), presented sequentially in the two placeholders (left/right). The order of presentation (left/right stimulus first) was balanced with respect to the longer/shorter duration of presentation. Participants were asked to indicate the location (left/right) of the longer duration blob. The shorter duration was always set to 600 ms, and the difference between shorter and longer durations was discretized in bins determined by the refresh rate of the monitor (100 Hz). The minimum duration difference was one single monitor refresh interval ($\approx$ 12 ms). The display screen was a 22 inches CRT monitor (screen resolution 1600×1200); viewing distance was 60 cm, controlled by means of a chin rest. Before the dual decision task, we measured individual sensitivities using a weighted up-down staircase procedure [4]. The purpose of this pre-test was to quickly obtain an estimate of the internal noise (σ) to adapt the range of stimuli in the dual-decision condition. The staircase procedure continued until 30 reversals were counted. The initial step size (the size of the decrease/increase of the difference between the two stimuli) was 4 refresh interval and was diminished 1 refresh after the second reversal. Stimuli in these pre-test measurements were presented only in the top placeholders. 5 participants run a version of the experiments where discrimination difficulties in the first and second decisions were drawn independently. The remaining 9 participants run a version of the experiment where we biased the probability of association between discrimination difficulties in the first and second decision. Specifically when the discriminability of the first stimulus was less than half the maximum possible value, there was a 0.7 probability that the discriminability of the second stimulus would be greater than half, and vice-versa. This was intended to encourage participants to make use of the rule, however we did not find any differences in performance across these two versions of the task and therefore data were pooled together in subsequent analyses. The number of trials were 300 and 500 for the version with biased and

random association between discrimination difficulties, respectively, to ensure balanced numbers of trial with transitions from easy to difficult decisions and viceversa.

2.2.2 Results

We applied the same exact analysis procedure used for the crossed-task experiment (see Main Text and Supplementary Methods). We found that decision order had no statistically significant effect in the control condition: 1.05, 95% CI [0.85, 1.31], $t(13) = 0.35$, $p = 0.74$; and that it had a significant effect in the dual-decision condition (Supplementary Figure 3B): the mean odds-ratio that participants responded correctly in the second decision relative to the first one was greater than one, 2.05, 95% CI [1.72, 2.51], $t(13) = 7.17$, $p = 7.12 \times 10^{-6}$. The size of this effect however was found to be larger than in the crossed-task experiment, $t(41) = 3.18$, $p = 0.003$.

When including the accuracy of the first decision as predictor, we found that also in this experiment participants were significantly more likely to choose 'right' in the second decision after a correct first decision than after an incorrect one: odds-ratio 2.36, 95% CI [1.69, 3.36], $t(13) = 4.72, p = 7.12 \times 10^{-6}$. The magnitude of this effect did not differ significantly from the crossed-task experiment, $t(41) = 0.58, p = 0.56$; the estimated Bayes factor (null/alternative) was 3.63, providing moderate support for the null hypothesis.

Moreover, by adding the discriminability of the first stimulus as a predictor, we found that it significantly influences the odds-ratio of choosing 'right' in the second decision: for each unitary increase in discriminability, the odds-ratio increases multiplicatively by a factor of 1.58, 95% CI [1.30, 1.95], $t(13) = 4.27, p = 9.13 \times 10^{-4}$, after a correct response and decreases by a factor of 0.74, 95% CI [0.57, 0.98], $t(13) = 2.07, p = 0.05$, after an incorrect response. There was no statistically significant difference in the size of both effects across the experiments: $t(41) = 0.62, p = 0.54$; and $t(41) = 0.32, p = 0.75$, respectively (the estimated Bayes factors, 3.55 and 4.02 respectively, providing moderate support for the null hypothesis).

Overall, these model-agnostic analyses indicate that although participants in the within-task experiment displayed a greater boost in performance (in decision 2 relative to decision 1), the extent by which the probability of choosing 'right' in the second decision changed as a function of first decision difficulty and accuracy did not differ significantly (in fact the Bayes factor provided evidence in favour of the null hypothesis of no difference between the two experiments). It thus remains unclear why participant in the within-task experiment achieved a greater boost in performance in the second decision: this may be due to the two decisions involving more similar judgments, or other differences in experimental procedures. For example, in the within-task experiment the range of stimuli was adjusted by running a staircase procedure, instead of a separate control condition, and on average it resulted in a slightly narrower range of stimuli when expressed in units of noise: the average range of stimuli in the within-task condition was $\pm 2.05\sigma$, whereas it was $\pm 2.67\sigma$ in the crossed-task experiment, $t(41) = 2.78, p = 0.008$.

Breaking down the trials according to the correctness of the first decision (Supplementary Figure 3C) reveals a pattern of deviations form the optimal Bayesian model very similar to what was observed in the crossed-task experiment, characterized by under-confidence after correct first decisions, albeit less evident, and with relatively higher-confidence errors. This pattern was again captured better by the discrete confidence model (compare data and model fit in Supplementary Figure 3C). However, in this case we find that the quality of model fit, as quantified by the AIC, did

not differ significantly between the discrete model and the biased-Bayesian model (Supplementary Figure 3D and E). The difference in AIC (summed over subjects) was 1.04 ± 11.26 in favour of the discrete model; the estimated Bayes factor (null/alternative) was 4.97, providing moderate evidence for the null hypothesis. These differences in quality of model fit, together with the greater boost in performance in the second decision, may indicate a more efficient use of confidence information in the within-task experiment. However, as explained above, there may be different explanations for this pattern that cannot be readily disentangled within the current set of results. On the one hand, it could indicated greater 'transmission fidelity' between internal states supporting decisions and confidence, i.e. a more precise and less noisy mapping between sensory evidence, confidence information and prior expectations when all decisions involve the same kind of sensory signals. On the other hand, it could indicate a greater heterogeneity in the strategies adopted by participants, since for this task the optimal strategy can be achieved effectively without needing additional information about noise or uncertainty. Since these difference computations may give rise to the same predicted behavioural outcome disentangling between them may require recording of neural activity in addition to modelling of behavioral data.

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